

PRACTICAL ASSIGNMENT SUBMISSION

NAME: SHRAVANI KHADILKAR

ROLL NO: 61

ASSIGNMENT: STUDENT CLASS

1. Source Code (Object-Oriented Implementation)

```
// Practical Assignment: Student Class Implementation
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

class Student {
private:
    string name;
    int rollNumber;
    float marks[3];
    float total;
    float percentage;

public:
    // Function to input student details
    void inputData(string sName, int rNo, float m1, float m2, float m3) {
        name = sName;
        rollNumber = rNo;
        marks[0] = m1;
        marks[1] = m2;
        marks[2] = m3;
        calculatePerformance();
    }

    // Function to calculate total and percentage
    void calculatePerformance() {
        total = marks[0] + marks[1] + marks[2];
        percentage = total / 3.0f;
    }

    // Function to display student record
    void displayData() const {
        cout << "===== " << endl;
        cout << "          STUDENT ACADEMIC REPORT          " << endl;
        cout << "===== " << endl;
        cout << left << setw(20) << "Student Name:" << name << endl;
        cout << left << setw(20) << "Roll Number:" << rollNumber << endl;
        cout << "-----" << endl;
        cout << left << setw(20) << "Subject 1 Score:" << marks[0] << " / 100" << endl;
    }
};
```

```

        cout << left << setw(20) << "Subject 2 Score:" << marks[1] << " / 100" << endl;
        cout << left << setw(20) << "Subject 3 Score:" << marks[2] << " / 100" << endl;
        cout << "-----" << endl;
        cout << left << setw(20) << "Total Marks:" << total << " / 300" << endl;
        cout << left << setw(20) << "Percentage:" << fixed << setprecision(2) << percent;
        cout << left << setw(20) << "Status:" << (percentage >= 40.0 ? "PASSED" : "FAILED");
        cout << "===== " << endl;
    }
};

int main() {
    Student student1;

    // Assigning student information and scores
    student1.inputData("SHRAVANI KHADILKAR", 61, 88.5, 92.0, 95.0);

    // Displaying output
    student1.displayData();

    return 0;
}

```

2. Program Execution Output

```

=====
                STUDENT ACADEMIC REPORT
=====
Student Name:      SHRAVANI KHADILKAR
Roll Number:      61
-----
Subject 1 Score:   88.5 / 100
Subject 2 Score:   92 / 100
Subject 3 Score:   95 / 100
-----
Total Marks:       275.5 / 300
Percentage:        91.83%
Status:            PASSED
=====

```